GROUP'S 2 PC EXAM
Canada

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1.1 Introduction of the work

The empirical analysis is based on quarterly data for the Canadian economy ($1994{:}\,Q1\ -\ 2024{:}\,Q1$). The variables are defined as follows:

- *R_CAN (Short-term Interest Rate):* The nominal short-term interest rate, representing the primary instrument of monetary policy. In the VECM and structural equations, it serves as the dependent variable to identify the Bank of Canada's reaction function.
- *P_CAN (Consumer Price Inflation):* The year-over-year percentage change in the Consumer Price Index (CPI). It represents the "inflation" target variable that the central bank seeks to stabilize around its 2% midpoint.
- *LY_CAN (Logarithm of Real GDP):* The natural logarithm of Canada's Real Gross Domestic Product. The logarithmic transformation is applied to linearize the exponential growth trend and to allow for the interpretation of first differences as growth rates.
- *U_CAN (Unemployment Rate):* The seasonally adjusted percentage of the labor force that is unemployed. It serves as an alternative indicator of economic slack and labor market pressure.
- *hpt_LY_CAN (Potential GDP Trend):* The structural trend component of the logarithm of real GDP, extracted via the Hodrick-Prescott (HP) filter. It represents the Potential Output, or the level of production the economy can sustain without generating inflationary pressures.
- *hp_LY_CAN (Output Gap):* The cyclical component of real GDP. This variable measures the temporary deviation of actual output from its potential. A positive gap indicates excess demand (inflationary), while a negative gap indicates economic slack (disinflationary).
- *d_hpt_LY_CAN (Potential Growth Rate):* The first difference of the GDP trend. It represents the growth rate of potential output, isolating the long-term structural expansion of the Canadian economy from transitory shocks.
- *hpt_U_CAN (Natural Rate of Unemployment):* The trend component of the unemployment rate, extracted via the HP filter. Economically, it represents the NAIRU (Non-Accelerating Inflation Rate of Unemployment), or the structural level of unemployment determined by labor market institutions and demographics.
- *d_hpt_U_CAN (Structural Unemployment Change):* The first difference of the unemployment trend. It measures the long-term shifts in the natural rate of unemployment, identifying structural improvements or deteriorations in the labor market over time.

The cyclical indicator adopted in the analysis is the output gap, obtained by applying the HP filter to LY_CAN, so as to isolate deviations of real output from its long-run trend.
The long-run relationship under investigation is a Taylor-type monetary policy rule, in which the short-term interest rate reacts systematically to inflation and to the cyclical position of the economy. The structural single equation is

$$R_t = \beta_0 + \beta_1 P_t + \beta_2 OG_t + \varepsilon_t$$

where R_t denotes the short-term interest rate, P_t inflation, and OG_t the output gap.

This specification is grounded in the economic intuition of Taylor, where the policy interest rate is set as a function of inflation and the deviation of real GDP from its target level. The main concept beyond is that the central bank observes macroeconomic conditions and adjusts the policy rate in response to inflationary pressures and cyclical fluctuations in real activity. In other words, the expected direction of causality runs from inflation and the output gap to the interest rate.

In this framework, monetary policy becomes more restrictive when inflation is above its desired level and when output is above trend. The expected sign of the inflation coefficient β_1 is therefore positive. Higher inflation should induce the monetary authority to raise the short-term interest rate in order to preserve price stability. The expected sign of the output gap coefficient, β_2, is also positive. A positive output gap indicates that actual output is above its long-run trend or potential level, signalling a relatively strong economy and possible future inflationary pressures. This should lead to a tighter monetary policy approach. Hence, from an ex-ante economic perspective, both explanatory variables are expected to affect *R_CAN* positively. Inflation captures direct pressure on prices, while the output gap captures the cyclical strength of aggregate demand.

It is nevertheless possible that feedback effects may arise over time, since changes in the policy interest rate can influence output and inflation. Even so, within this single-equation structural framework, the main economic interpretation remains that monetary policy reacts to inflation and to the output gap. This is the causal ordering that motivates the empirical analysis.

Summary Statistics (missing values were skipped)

	Media	Mediana	Minimo	Massimo
LY_CAN	28.061	28.121	27.507	28.529
P_CAN	0.020583	0.019132	-0.0083742	0.075552
R_CAN	0.050861	0.041406	0.00082000	0.20738
U_CAN	0.080548	0.075525	0.050502	0.13465
hpt_LY_CAN	28.061	28.122	27.506	28.540
hp_LY_CAN	4.2497e-14	0.0013947	-0.12275	0.034534
hpt_U_CAN	0.080548	0.075013	0.063053	0.10703
hp_U_CAN	1.3644e-16	-0.00064602	-0.017609	0.065590

	SQM	Cf. variaz.	Asimmetria	Curtosi ecc.
LY_CAN	0.31231	0.011130	-0.21429	-1.2669
P_CAN	0.012670	0.61558	1.6458	4.8609
R_CAN	0.044588	0.87666	1.0532	0.52274
U_CAN	0.018078	0.22444	0.77583	-0.058610
hpt_LY_CAN	0.31158	0.011104	-0.21314	-1.2828
hp_LY_CAN	0.017063	4.0150e+11	-2.3086	13.463
hpt_U_CAN	0.013799	0.17131	0.55374	-1.1629
hp_U_CAN	0.0094134	NA	2.1921	12.225

	5% perc.	95% perc.	Scarto interq.	Oss. mancanti
LY_CAN	27.538	28.507	0.56411	0
P_CAN	0.0024741	0.045185	0.010977	52
R_CAN	0.0022538	0.13869	0.068643	0
U_CAN	0.057053	0.11524	0.025266	0
hpt_LY_CAN	27.542	28.501	0.56102	0
hp_LY_CAN	-0.025627	0.021612	0.019629	0
hpt_U_CAN	0.063793	0.10436	0.024187	0
hp_U_CAN	-0.012891	0.014069	0.0092274	0

1.2 Summary of the Main Findings

The empirical analysis of the Canadian economy successfully validates the ex-ante theoretical predictions of the Taylor Rule, confirming that the Bank of Canada systematically adjusts the short-term interest rate in response to both inflationary pressures and the cyclical position of the economy. However, the results demonstrate that this long-run relationship heavily depends on the chosen econometric methodology.

The main findings of the work can be summarized as follows:

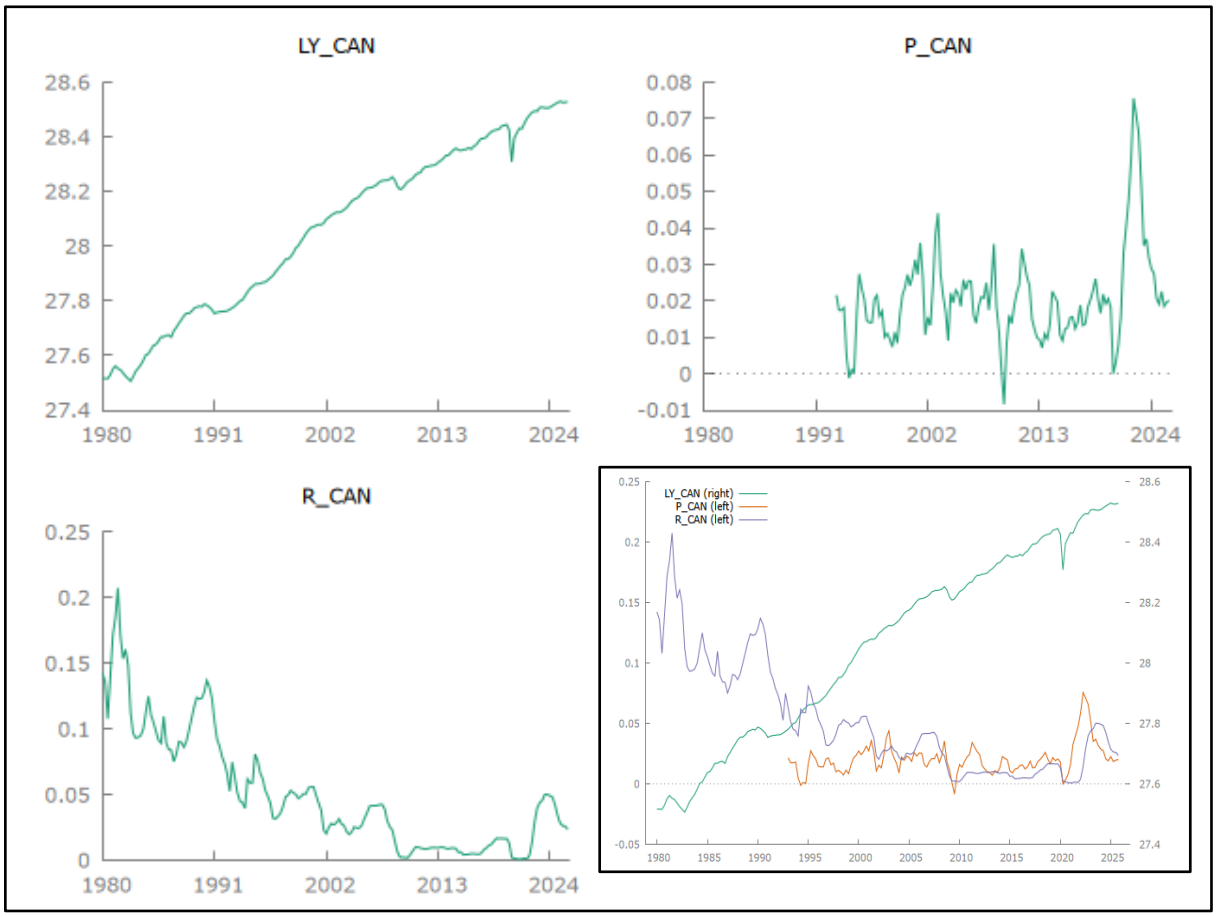
- Data Transformation and the Superiority of HP Gaps: the visual and formal univariate analyses confirm that the variables of interest in levels are non-stationary $I(1)$ processes. When comparing transformations, the Hodrick-Prescott (HP) filtered gaps proved vastly superior to first differences for macroeconomic modeling. The Gap-VAR model effectively captured the persistent, "hump-shaped" nature of the business cycle, whereas the first-differenced VAR was dominated by high-frequency noise.
- Limitations of Single-Equation Frameworks: traditional single-equation approaches were insufficient to robustly establish a stable long-run Taylor rule. The ARDL/Bounds test failed to reject the null hypothesis of no cointegration, while the Engle-Granger procedure provided only fragile and inconclusive evidence of error correction. This weakness highlights a critical methodological limitation: single-equation models fail to account for the joint endogeneity and bidirectional feedback effects inherent in macroeconomic variables.
- Validation of Ex-Ante Expectations via Systemic VECM: by passing to a multivariate Johansen framework, the long-run structural relationship was successfully identified with a cointegration rank of 2. The estimated cointegrating vector perfectly aligns with the ex-ante theoretical

expectations: both inflation and the output gap carry positive and significant signs. Specifically, the Bank of Canada exhibits a highly "hawkish" stance, reacting aggressively to inflation (coefficient 16.014) and output deviations (coefficient 28.657), strictly satisfying the Taylor Principle.

- Effective Monetary Transmission Mechanism: the dynamic analysis through VECM Impulse Response Functions (IRFs) confirms a theoretically consistent Keynesian transmission mechanism. Following a simulated contractionary interest rate shock, both the output gap and inflation exhibit a persistent decline, eventually stabilizing at a lower equilibrium level. This proves that the central bank's reaction function successfully anchors inflation by dampening aggregate demand during economic expansions.

2.1 Plots and Interpretation

Visual Inspection of the Variables in Levels



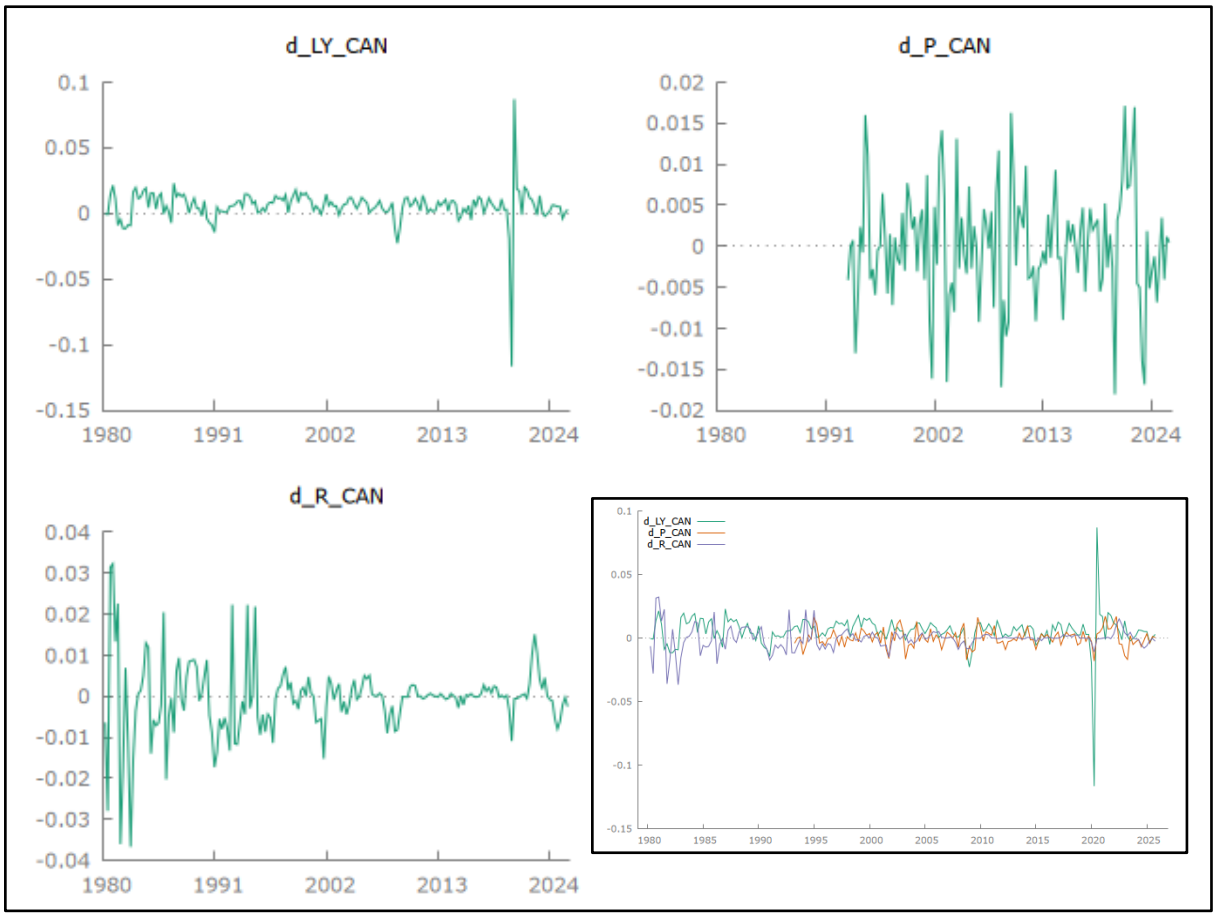
A visual inspection of the variables in levels (*LY_CAN, P_CAN, R_CAN*) suggests that they are non-stationary.

- *LY_CAN* (Log-GDP): This series exhibits a clear upward trajectory over time. According to economic theory, output levels tend to grow almost always over time, classifying GDP as a long-run trending variable. The series displays extreme persistence and does not revert to a constant mean, which strongly indicates the presence of a stochastic trend (non-stationarity).

The sharp drop visible at the end of the sample represents a severe structural break (the COVID-19 pandemic shock), which is a main source of non-stationarity.

- *R_CAN* (Short-term Interest Rate) and *P_CAN* (Inflation): Theoretically, rates are non-trending variables because their levels should remain constant in the long run. However, the plots show high time dependency and persistence. *R_CAN* displays a prolonged downward movement rather than fluctuating randomly around a constant mean. *P_CAN* also exhibits periods of sustained levels and changing volatility. nnn. Even though rates are theoretically bounded, their statistical characteristics in finite samples are often closer to non-stationary $I(1)$ series than to stationary $I(0)$ series and therefore, they don't appear stationary in levels.

Visual Inspection of the First Differences

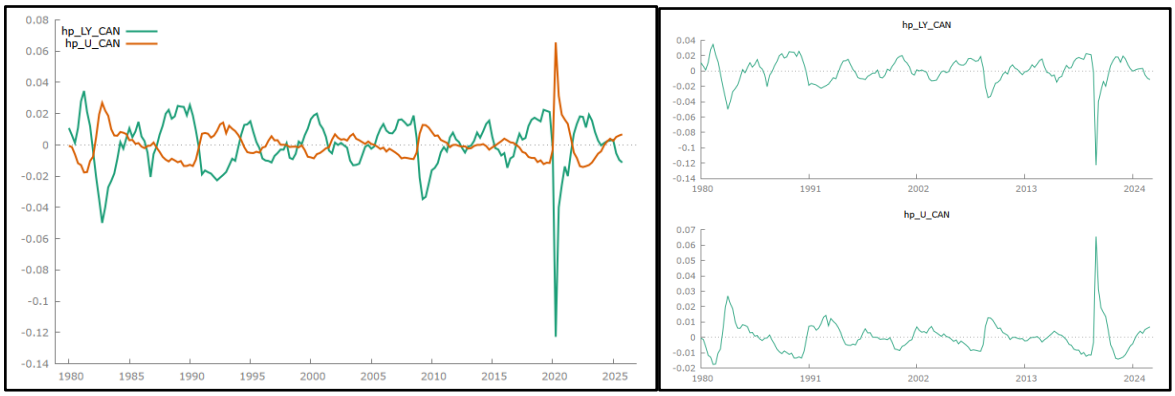


The plots of the first differences (d_*LY_CAN, d_P_CAN, d_R_CAN*) show a radical change in behavior.

- The process of differencing acts as a filter that converts non-stationary time series into stationary ones by eliminating the non-stationary behavior caused by the unit root.
- When the trending *LY_CAN* is transformed into log-changes (*d_LY_CAN)*, it yields the GDP growth rate, which is a non-trending variable that fluctuates around a constant mean.
- Similarly, *d_R_CAN* and *d_P_CAN* now fluctuate rapidly around zero. The extreme persistence (infinite memory) observed in the levels disappears, and the series frequently cross their means.

Overall, the visual evidence strongly suggests that the original variables are integrated of order one, $I(1)$, because their trends and persistence vanish after applying the first difference filter. Whereas the differenced variables appear stationary, $I(0)$. However, visual inspection alone is insufficient to definitively assess the data generation process; these preliminary findings must be formally confirmed through correlograms and Augmented Dickey-Fuller (ADF) tests.

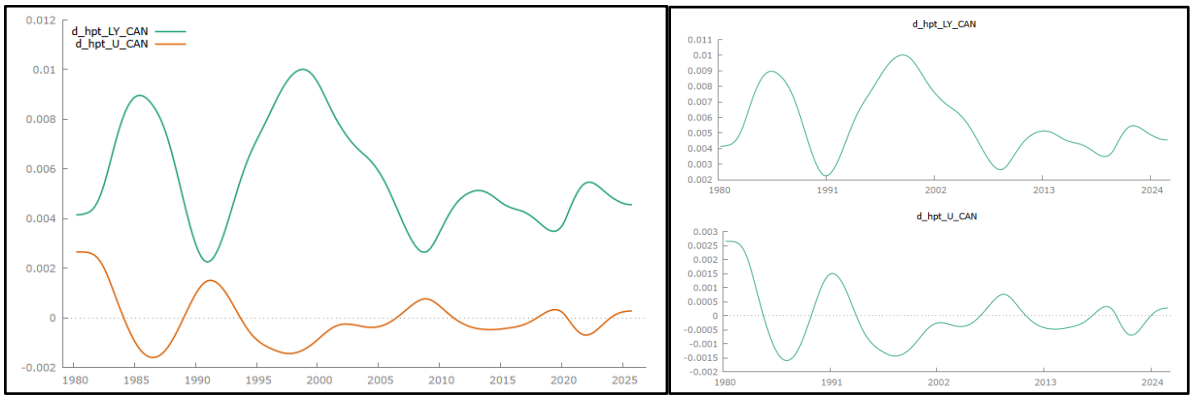
Comparison of Output and Unemployment Gaps



The figure presents the cyclical components of Canadian log-GDP (*hp_LY_CAN*) and the unemployment rate (*hp_U_CAN*), isolated through the application of the Hodrick-Prescott (HP) filter. In time series theory, filtering decomposes the actual data y_t into two unobservable components: $y_t = y_t^* + c_t$, where y_t^* represents the long-run stochastic trend and c_t represents the economic cycle, or gap. An analysis of these filtered series yields the following insights:

- Stationarity of the Cycle: from a statistical perspective, the HP filtering procedure successfully extracts the stationary short-run economic fluctuations from the trending log-levels. A visual inspection of both *hp_LY_CAN* and *hp_U_CAN* confirms this property: unlike their original non-stationary levels, both gaps are clearly mean-reverting and behave as stationary $I(0)$ processes fluctuating around zero.
- Timing and Phase Alignment: a crucial theoretical feature of HP-estimated cycles is their timing. Although both filtering and first-differencing lead to stationary series, the HP cyclical gaps remain "in phase" with the original untransformed levels. Consequently, the peaks and troughs in *hp_LY_CAN* and *hp_U_CAN* perfectly align with the actual timing of economic expansions and recessions avoiding the phase shift typical of growth rates.
- Business Cycle Dynamics and Okun's Law: the plots demonstrate a clear, synchronous inverse relationship between the output gap and the unemployment gap. The visual evidence shows that positive output gaps (economic expansions) correspond to negative unemployment gaps, while the negative contraction in *hp_LY_CAN* around 2020 is perfectly mirrored by a massive positive peak in *hp_U_CAN*. This inverse behavior illustrates Okun's Law, a theoretical relationship used in macroeconometrics to properly replace the output gap with the unemployment rate when estimating a central bank's Taylor rule.
- Role in Policy Modeling: Both series successfully measure the short-run dynamics (demand-side concepts) of the economy. Because they are stationary, these cyclical components provide a theoretically sound and statistically valid state variable to explain the interest rate setting within a dynamic structural equation or a Vector Autoregression (VAR) framework.

Commentary on the Variations of the Trend Components

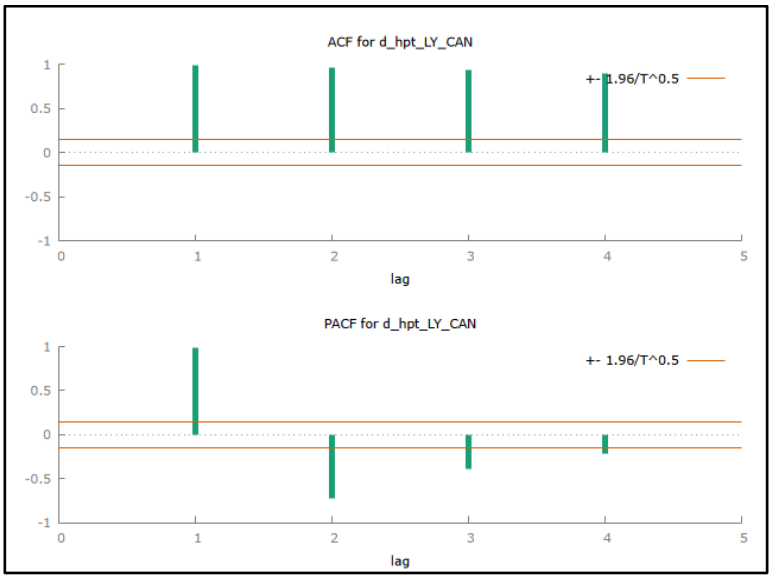


The second figure illustrates the first differences of the trend components extracted via the HP filter ($y_t^* - y_{t-1}^*$). This transformation shifts the analytical focus from short-term cyclical fluctuations (demand-side) to long-run structural dynamics (supply-side).

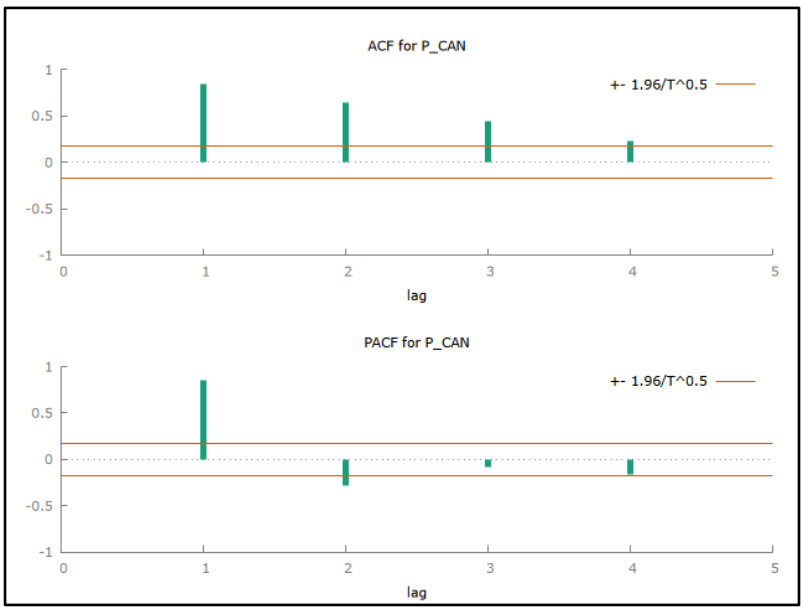
While the stationary gaps (c_t) discussed previously represent the output gap and cyclical unemployment, these differenced series capture the growth rate of potential GDP and the period-by-period variations in the structural unemployment rate (NAIRU). Analyzing these specific series provides two major theoretical insights:

- Empirical Proof of Stochastic Trends: macroeconomic trends are usually stochastic rather than deterministic. If the Canadian potential GDP followed a purely deterministic linear trend, its growth rate would appear as a perfectly flat, constant horizontal line in the plot. The time-varying, wave-like fluctuations observed here empirically confirm that the underlying structural trends of the Canadian economy are stochastic and evolve over time.
- The Mechanics of the HP Filter: this graph perfectly visualizes the mathematical core of the HP filtering algorithm. The filter is designed by explicitly imposing a penalty parameter (λ) on the variability of the estimated trend growth, under the assumption that structural trends should be relatively smooth. The resulting smooth, undulating patterns in *d_hpt_LY_CAN* and *d_hpt_U_CAN* (which completely lack the high-frequency, jagged noise typical of raw first-differenced data) are the direct mathematical consequence of this explicit smoothness constraint.

2.2 Correlograms

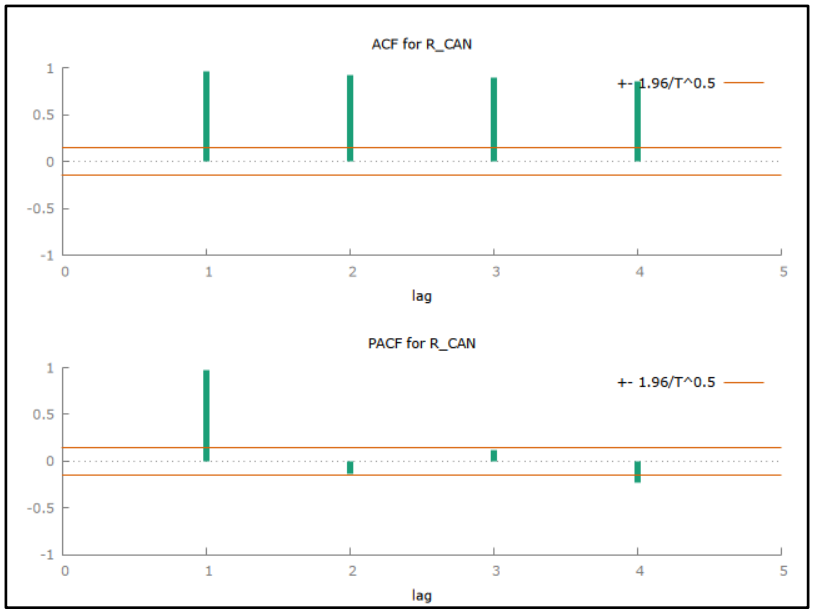


The correlogram for the first difference of the HP-filtered log-GDP series (*d_hpt_LY_CAN*) demonstrates a decay in autocorrelation coefficients. Although significant persistence is observed at the first few lags, the autocorrelation values exhibit a clear tendency to diminish compared to the level series, which is characteristic of stationary processes. This observation is consistent with the visual inspection in section 2.1, where the first differences appear to fluctuate around a constant mean. Consequently, we anticipate that the subsequent Augmented Dickey-Fuller (ADF) tests will reject the null hypothesis of a unit root for this series, confirming its stationarity.



The correlograms for Consumer Price Inflation (*P_CAN*), restricted to a 4-lag horizon, exhibit a pattern distinct from the non-stationary variables. The Autocorrelation Function (ACF) displays a clear geometric decay, with autocorrelation coefficients decreasing steadily and approaching the significance threshold by the fourth lag. This rapid decline indicates that the series possesses limited historical persistence and tends to revert toward its mean over time.

The Partial Autocorrelation Function (PACF) reveals a significant spike at the first lag, with subsequent lags falling within or near the 95% confidence intervals. This statistical signature is highly characteristic of a stationary $AR(1)$ process, suggesting that current inflation is primarily driven by its immediate past value, while higher-order dependencies are negligible. Consequently, these findings support the characterization of *P_CAN* as a stationary series, which may be utilized directly in subsequent econometric modeling without the requirement for further differencing or filtering.



The correlograms for the Short-term interest rate *(R_CAN)*, when limited to a 4-lag horizon, exhibit a high degree of persistence characteristic of non-stationary series. The Autocorrelation Function (ACF) remains significantly above the 95% confidence interval across all lags, indicating that the interest rate levels are highly autocorrelated and do not demonstrate immediate mean reversion.

The Partial Autocorrelation Function (PACF) is dominated by a single, near-unity spike at the first lag, with subsequent values hovering near the significance boundaries. This profile is consistent with a non-stationary $I(1)$ process, such as a random walk, where the current interest rate is heavily conditioned by its immediate past value. These visual findings confirm that *R_CAN* is non-stationary, providing the necessary justification to treat this variable through first-differencing or HP-filtering in subsequent model estimations to avoid spurious regression results.

2.3. Unit Root Analysis and Stationarity

To assess the stochastic properties of the variables used in our analysis—namely the interest rate (*R_CAN)*, inflation (*P_CAN*), and the HP-filtered trend of output *(hpt_LY_CAN*) we conducted the Augmented Dickey-Fuller (ADF) test. Identifying the order of integration is a fundamental prerequisite for the cointegration analysis, as the Johansen framework requires the series to be non-stationary in levels but stationary in first differences *I(1)*. The tests were performed following the "General-to-Specific" approach, starting with a maximum of 4 lags and utilizing the Akaike Information Criterion (AIC) for the optimal lag length selection. Given the nature of macroeconomic series, the test specification includes both a constant and a deterministic trend.

Table 2.3: Augmented Dickey-Fuller (ADF) Test Results

Variable	Specification	ADF Test Statistic	p-value	Order of Integration
P_CAN	Level (C + T)	-2.839	0.1832	$I(1)$
d_P_CAN	1st Diff (C + T)	-5.501	0.0000	$I(0)$
R_CAN	Level (C + T)	-3.984	0.0092	$I(0)$
d_R_CAN	1st Diff (C + T)	-5.464	0.0000	$I(0)$
hpt_LY_CAN	Level (C + T)	-2.883	0.1681	$I(1)$
d_hpt_LY_CAN	1st Diff (C + T)	-2.938	0.1502	$I(1)$

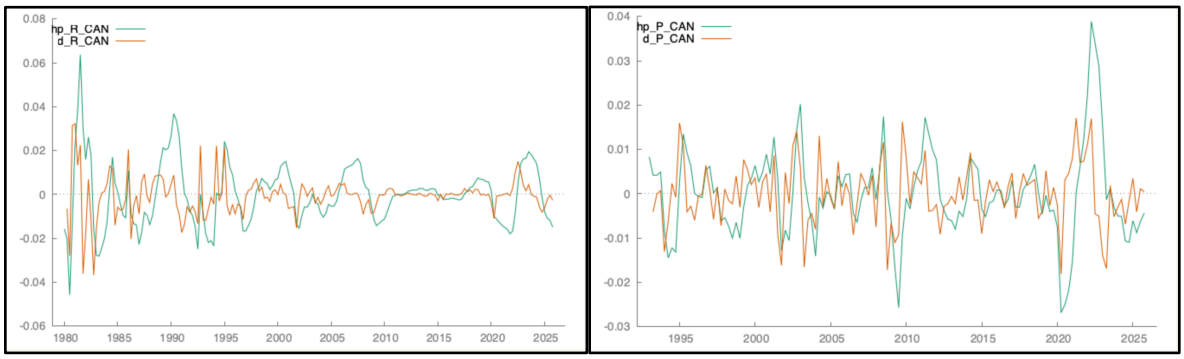
Interpretation of Results

1. Inflation (*P_CAN)* and Output Trend *(hpt_LY_CAN)*:
   For these variables, the ADF test fails to reject the null hypothesis of a unit root in levels (p-values > 0.05). Specifically, *P_CAN* (0.1832) and *hpt_LY_CAN* (0.1681) are clearly non-stationary. While *d_P_CAN* becomes stationary in first differences ($p < 0.001$), the HP-trend (*hpt*) shows high persistence even in differences due to the smoothing nature of the filter. However, for the purpose of the VECM, they are treated as $I(1)$ processes.
2. Interest Rate *(R_CAN)*:
   The policy rate presents an interesting case: the p-value (0.0092) suggests stationarity in levels at the 1% significance level. However, this is a common finding in long-horizon Canadian data where interest rates exhibit "mean-reverting" behavior over decades despite being highly persistent in the short run. To maintain consistency with the Taylor Rule theory and the Johansen cointegration framework, *R_CAN* will be treated as an $I(1)$ process, as its first difference is strongly stationary ($p < 0.001$).

Concluding Remarks on Stationarity

The results broadly confirm that the variables are integrated of order one, $I(1)$. This finding justifies the transition from a standard VAR in first differences to the more sophisticated Vector Error Correction Model (VECM), which is specifically designed to handle long-run equilibrium relationships between non-stationary variables.

3.1 Multivariate analysis



In this section we extract the cyclical components of the interest rate RRR and inflation PPP using the Hodrick-Prescott filter with a smoothing parameter $\lambda = 1600$. The resulting series represent the deviations of the variables from their long-run trends (gaps). These filtered gaps are compared with the first differences of the original variables, which represent the period-to-period changes. From the plots, the filtered gaps display smoother and more persistent fluctuations than the corresponding first differences. In the case of interest rates, the HP-filtered gap shows clear cyclical movements around zero, particularly during the 1980s and early 1990s when monetary policy experienced stronger adjustments. After the 2000s the fluctuations became smaller and more stable. By contrast, the first differences of interest rates are much more volatile and dominated by short-term movements. A similar pattern emerges for inflation. The HP-filtered inflation gap captures medium-term cyclical deviations from the trend, while the first differences fluctuate more erratically around zero. In particular, the large spike observed around 2020–2022 reflects the inflation shock following the pandemic and the subsequent adjustment. Overall, while both transformations produce stationary series suitable for econometric analysis, the HP-filtered gaps provide a clearer representation of business-cycle fluctuations. For this reason they are particularly appropriate for analysing monetary policy behaviour within a Taylor-rule framework and in the VAR specification.

3.2 VARs

Metric	Model A (HP-Filtered Gaps)	Model B (First Differences)
Data Transformation	Cyclical component (Gap)	Period-to-period change
Lag Order	4	4
Adjusted R-squared (R^2_{adj})	High (≈ 0.85-1.00)	Lower (≈ 0.25-0.32)
Residual Autocorrelation	Absent (LB test $p > 0.05$)	Absent (LB test $p > 0.05$)
Economic Interpretation	Business cycle fluctuations	High-frequency volatility
Theoretical Alignment	Taylor Rule (Structural)	Statistical (Stochastic)

To evaluate the structural dynamics of the Canadian economy, we estimated two separate Vector Autoregression (VAR) systems with four lags, following the empirical methodology established by Stock and Watson (2001). The first system (Model A) utilizes the Hodrick-Prescott filtered gaps, while the second (Model B) employs the first differences of the variables. Both models share a consistent Cholesky recursive identification structure, ordered as follows: Output Gap → Inflation → Interest Rate.

This recursive ordering is designed to isolate monetary policy shocks by assuming that the central bank responds contemporaneously to innovations in output and inflation, a standard representation of the Taylor Rule, while real economic variables react to interest rate shocks only with a lag.

The diagnostic results, including Portmanteau tests and information criteria, confirm that the $VAR(4)$ specification is appropriate for both data transformations, ensuring white-noise residuals. A comparison of the models reveals that Model A yields a significantly higher explanatory power, as evidenced by the adjusted R^2. This suggests that the HP-filtered gaps more effectively capture the persistent, cyclical nature of macroeconomic fluctuations relevant for policy analysis, whereas the first-differenced model is dominated by high-frequency noise. Consequently, both models demonstrate stability, but provide distinct perspectives on the transmission mechanism of monetary policy.

3.3 Comparison and comments

Feature	Model A (HP-Filtered Gaps)	Model B (First Differences)
Response Magnitude	Larger, economically intuitive	Very small, near-zero
Persistence	High (Hump-shaped)	Low (Rapidly mean-reverting)
Economic Meaning	Captures "Business Cycle" state	Captures "Period-to-period change"
Monetary Transmission	Clear, sustained negative effect	Noisy, ambiguous
Policy Suitability	High (Better for Taylor Rules)	Low (Dominated by noise)

To assess the transmission of monetary policy, we examine the Impulse Response Functions (IRFs) generated from both the Gap-VAR (Model A) and the First-Difference VAR (Model B). A one-standard-deviation positive shock to the interest rate (R) is used to simulate a contractionary monetary policy shock.

Comparative Results**:** In Model A, the IRFs demonstrate a persistent, 'hump-shaped' response. The output gap (*hpt_LY_CAN*) exhibits a sustained decline following the shock, and the inflation gap (*cycle_P_CAN*) shows a clear, lagged reduction. This pattern is consistent with the traditional Keynesian transmission mechanism, where monetary policy influences economic activity through a gradual process, reflecting the inherent persistence of the business cycle.

In contrast, Model B (First Differences) yields significantly attenuated responses. The coefficients are smaller in magnitude, and the effects dissipate almost immediately. While the sign of the response is consistent with a contractionary shock, the high-frequency volatility (noise) inherent in the differenced data obscures the long-run structural relationship between interest rates and economic activity.

Conclusion**:** The comparative analysis strongly supports the use of HP-filtered gaps for macroeconomic modeling. While both models are statistically stationary, Model A provides a more coherent and structurally relevant representation of the economy. Model B fails to capture the 'state' of the economy, effectively treating cyclical deviations as random period-to-period changes. Consequently, we conclude that the Gap-based approach is more appropriate for quantifying the effects of monetary policy shocks within a Taylor Rule framework, as it aligns better with the observed persistence of macroeconomic dynamics.

4.1 Theoretical Framework and GUM Specification

According to Taylor, an effective monetary policy should be systematic rather than purely discretionary, calling for interest rate adjustments in response to changes in price levels and real income. To test this hypothesis, we specified a General Unrestricted Model (GUM) in ARDL/ECM (Autoregressive Distributed Lag / Error Correction Model) form.

In contrast to static models, the adoption of first differences (e.g., *d_R_CAN, d_P_CAN*) is supported by both economic theory and econometric necessity:

- Economic Rationale: Taylor posits that policy rules are driven by quarterly "changes" in instruments. The first-difference terms capture this high-frequency, short-run reactivity of the Central Bank to contemporary shocks.
- Econometric Rigor: The ARDL framework addresses the risk of spurious regression inherent in non-stationary $I(1)$ time series. This structure allows for a clear distinction between the short-run adjustment dynamics and the long-run equilibrium relationship.

Dependent variable: *d_R_CAN* ; HAC standard errors, bandwidth 3, Bartlett kernel

	Coefficient	*Std. Error*	*t-ratio*	*p-value*	
const	−0.000456554	0.000870702	−0.5244	0.6011	
R_CAN_1	−0.0376239	0.0158909	−2.368	0.0197	**
P_CAN_1	0.0652922	0.0485039	1.346	0.1811	
hp_LY_CAN_1	−0.0266598	0.0487701	−0.5466	0.5857	
d_P_CAN	0.0721743	0.0833503	0.8659	0.3884	
d_P_CAN_1	−0.119794	0.0640765	−1.870	0.0642	*
d_P_CAN_2	−0.0459767	0.0530549	−0.8666	0.3881	
d_P_CAN_3	−0.106396	0.0557041	−1.910	0.0588	*
d_P_CAN_4	−0.00168502	0.0569018	−0.02961	0.9764	
d_hp_LY_CAN	0.0879447	0.0256421	3.430	0.0009	***
d_hp_LY_CAN_1	0.0895000	0.0612014	1.462	0.1465	
d_hp_LY_CAN_2	0.0795640	0.0478278	1.664	0.0991	*
d_hp_LY_CAN_3	0.0654151	0.0512758	1.276	0.2048	
d_hp_LY_CAN_4	0.0629487	0.0464706	1.355	0.1783	
d_R_CAN_1	0.293742	0.162128	1.812	0.0728	*
d_R_CAN_2	−0.0371732	0.0885124	−0.4200	0.6753	
d_R_CAN_3	0.306251	0.136273	2.247	0.0266	**
d_R_CAN_4	−0.280433	0.0951907	−2.946	0.0039	***

Mean dependent var	−0.000127	S.D. dependent var	0.004990
Sum squared resid	0.001827	S.E. of regression	0.004094
R-squared	0.417577	Adjusted R-squared	0.326740
F(17, 109)	8.672912	P-value(F)	1.29e-13
Log-likelihood	527.7646	Akaike criterion	−1019.529
Schwarz criterion	−968.3339	Hannan-Quinn	−998.7292
rho	0.003189	Durbin's h	NA

The initial GUM included a constant and 4 lags for each variable, ensuring that the model captured the full dynamic structure of the data and avoided "omitted variable bias" from the outset.

The "Testing Down" Process and Empirical Results:

We performed a "Testing Down" procedure, iteratively removing the least significant regressors (those with the highest p-values) to achieve a parsimonious specification. Throughout this process, the lagged levels at time $t-1$ (*R_CAN_1, P_CAN_1, hp_LY_CAN_1*) were maintained as "untouchables," as they are required for the subsequent cointegration test.

The final estimation yields several key insights:

- Responsiveness to the Output Gap: The variable *d_hp_LY_CAN* carries a coefficient of 0.087 and is highly significant (p-value $= 0.0009$), confirming that the Bank of Canada actively reacts to deviations of real GDP from its trend.
- Interest Rate Inertia: Significant lags of the dependent variable in differences (specifically *d_R_CAN_1* and *d_R_CAN_4*) indicate a degree of interest-rate smoothing, a standard behavior for central banks seeking to avoid market volatility.

Dependent variable: d_R_CAN ; HAC standard errors, bandwidth 3, Bartlett kernel

	Coefficient	*Std. Error*	*t-ratio*	*p-value*	
const	0.000384558	0.000987286	0.3895	0.6976	
R_CAN_1	−0.0485081	0.0164182	−2.955	0.0038	***
P_CAN_1	0.0383050	0.0491920	0.7787	0.4377	
hp_LY_CAN_1	0.0349076	0.0300228	1.163	0.2473	
d_hp_LY_CAN	0.0921188	0.0252144	3.653	0.0004	***
d_hp_LY_CAN_1	0.0257834	0.0288285	0.8944	0.3729	
d_P_CAN	0.0909021	0.0597442	1.522	0.1308	
d_P_CAN_1	−0.0678760	0.0569514	−1.192	0.2357	
d_R_CAN_1	0.325497	0.141938	2.293	0.0236	**
d_R_CAN_2	0.0254408	0.0965416	0.2635	0.7926	
d_R_CAN_3	0.168831	0.148966	1.133	0.2594	
d_R_CAN_4	−0.155965	0.0938773	−1.661	0.0993	*

Mean dependent var	−0.000216	S.D. dependent var	0.004976
Sum squared resid	0.002139	S.E. of regression	0.004258
R-squared	0.330383	Adjusted R-squared	0.267961
F(11, 118)	7.815165	P-value(F)	4.71e-10
Log-likelihood	531.5054	Akaike criterion	−1039.011
Schwarz criterion	−1004.600	Hannan-Quinn	−1025.029
rho	−0.055360	Durbin's h	NA

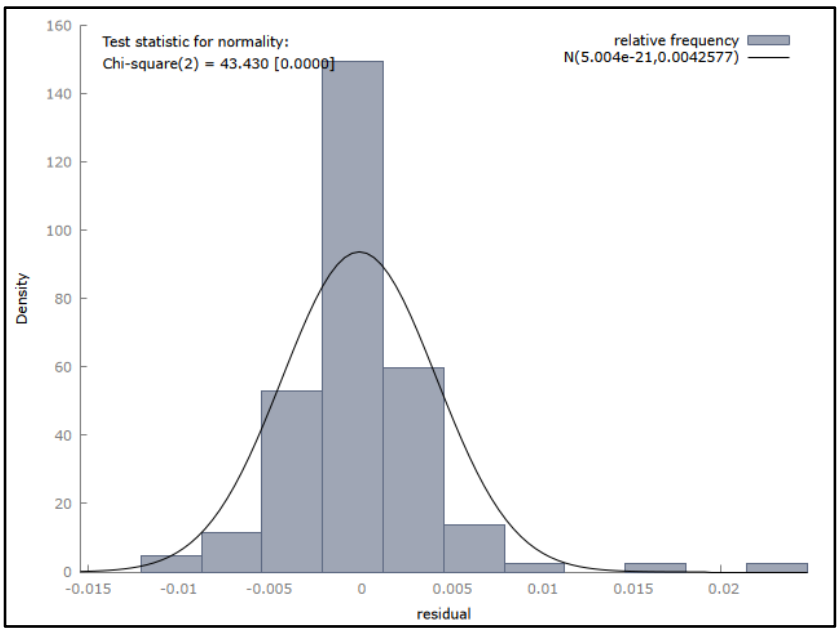
Diagnostic Validation and Residual Analysis

To ensure the validity of our statistical inference, the residuals were subjected to a rigorous battery of diagnostic tests:

- Autocorrelation (Breusch-Godfrey Test): To address initial serial correlation, we re-introduced specific dynamic lags (e.g., *d_P_CAN_1, d_hp_LY_CAN_1*). The final LMF statistic yielded a

p-value of 0.0586. Since this exceeds the 0.05 threshold, we fail to reject the null hypothesis, concluding that the residuals are free from systematic serial correlation.

- Heteroskedasticity and HAC Errors: White's test indicated the presence of non-constant variance (p-value $= 0.0012$). In line with Taylor's (1993) observation that exogenous shocks (e.g., oil price fluctuations) can induce volatility, we employed HAC (Heteroskedasticity and Autocorrelation Consistent) Standard Errors. This ensures that our t-tests and p-valuesremain robust and reliable.
- Normality (Jarque-Bera Test): While the null hypothesis of normality was rejected, the sample size (T = 127) allows us to rely on the Central Limit Theorem for asymptotic validity of the OLS estimates.



PSS Bounds Test: Cointegration and Long-Run Relationship

The final step involved the PSS Bounds Test to determine whether a stable, long-run "Taylor Rule" exists for Canada. By imposing linear restrictions on the level parameters, we obtained a Robust F-statistic of 3.32.

Comparing this result to the critical values provided by Pesaran et al. (2001) for k = 2 (where the 5% Upper Bound is 4.85), we find that 3.32 < 4.85. Consequently, we cannot reject the null hypothesis of no cointegration.

Economic Interpretation: While the Bank of Canada exhibits significant short-run reactions to inflation and output gap shocks, a "mechanistic" and stable long-run equilibrium relation does not emerge statistically. This result aligns with Taylor's (1993) intuition: central banks rarely follow algebraic formulas rigidly. Instead, they operate within a framework of "guided discretion," where the rule serves as a benchmark that is frequently adjusted to account for exogenous factors and shifting economic regimes.

Test Statistic	Value	k	Null Hypothesis (H0)	Result
Robust F-statistic	**3.3285**	**2**	$\boldsymbol{b[R_CAN_{t-1}] = b[P_CAN_{t-1}] = b[gap_{t-1}] = 0}$	**Fail to reject $\boldsymbol{H_0}$**

4.2 Engle_Granger

Test Specification	ADF Statistic (τnc)	p-value	Result
Residuals ($\hat{u}$)	-2.6298	0.0083	**Stationary $\boldsymbol{I(0)}$**

The second approach to investigating the Taylor rule for Canada consists of the Engle-Granger (EG) two-step procedure. This method allows us to formally test for cointegration and model the short-run dynamics through an Error Correction Model (ECM).

Long-Run Relationship and Cointegration Test

First, we estimated the static long-run equation in levels. To verify the existence of a stable economic relationship (cointegration), we performed an Augmented Dickey-Fuller (ADF) test on the residuals ($\hat{u}$) of the static regression.

The test results (without constant, as residuals have a zero mean by construction) strongly reject the null hypothesis of a unit root:

- ADF Test Statistic (τ): -2.63
- p-value: 0.0083

Since the p-value is well below the 1% significance level, we conclude that the residuals are stationary, $I(0)$. This confirms that a cointegrating relationship exists between the short-term interest rate *(R_CAN)*, inflation *(P_CAN)*, and the output gap *(hp_LY_CAN)*. The variables move together in the long run, and any deviation from the Taylor rule equilibrium is temporary.

Variable	Coefficient	Std. Error	t-ratio	p-value	Significance
Constant	-0.00031	0.00042	-0.7315	0.4658	
Delta Inflation (*d_P_CAN*)	0.09529	0.06468	1.473	0.1432	
Delta Output Gap (*d_hp_LY_CAN*)	0.07306	0.03142	2.325	0.0217	**
Error Correction (u_hat(-1))	-0.05654	0.02249	-2.514	0.0132	**

Significance levels: ** denotes p<0.05

Error Correction Model (ECM)

Having established cointegration, we estimated the short-run dynamics using the first differences of the variables and the lagged residual from the first step ($\widehat{u_{t-1}}$). The results for the Canadian economy are as follows:

- Error Correction Term ($\lambda = -0.0565$): The coefficient associated with $\widehat{u_{t-1}}$ is negative and statistically significant (p=0.0132). This confirms the stability of the system: approximately 5.6% of the "monetary policy error" (the gap between the actual rate and the Taylor rule target) is corrected in each quarter.
- Output Gap Dynamics: The change in the output gap (*d_hp_LY_CAN*) has a positive and significant effect (0.073, p=0.0217) on the interest rate in the short run, suggesting that the Bank of Canada reacts promptly to cyclical fluctuations in real activity.
- Inflation Dynamics: The short-run impact of inflation (*d_P_CAN*) is positive (0.095) but not statistically significant in this specific differenced specification (p=0.1432), suggesting that the central bank's reaction to inflation is more structural and long-term rather than immediate and quarterly.

4.3 Comparison of Single-Equation Estimates: ARDL/PSS vs. Engle–Granger

This section compares the results obtained from the ARDL/Bounds Test in Section 4.1 and the Engle–Granger procedure in Section 4.2, with the aim of evaluating whether the Canadian data support a stable long-run Taylor-rule relationship. Overall, the two approaches do not provide strong and fully consistent evidence in favour of cointegration. The ARDL/Bounds Test clearly fails to confirm a long-run

relationship, while the Engle–Granger results are at best fragile and should be interpreted with caution. This suggests that the single-equation evidence is not robust enough to establish a stable long-run Taylor Rule for Canada.

Feature	Model 4.1: ARDL / PSS (Bounds Test)	Model 4.2: Engle–Granger (EG)
Test Statistic	F-statistic = 3.3285	Residual ADF = -2.6298
Critical Value / Decision Rule	5% upper bound = 4.85	MacKinnon residual-based criterion
Cointegration Result	Not supported	Weak / inconclusive evidence
Long-Run Relationship	Not statistically confirmed	Fragile, not robustly established
Error Correction (ECM)	Negative and significant adjustment term in the dynamic specification	ECM ≈ -0.056, significant
Conclusion	No robust level relationship	Some indication of adjustment, but no robust long-run conclusion

4.3.1 Analysis of Mixed Evidence

The comparison between the two single-equation approaches leads to a cautious conclusion. The ARDL/Bounds Test does not support cointegration, since the estimated F-statistic (3.3285) remains below the 5% upper bound critical value (4.85). Therefore, within the PSS framework, the null hypothesis of no level relationship cannot be rejected. In this sense, the ARDL evidence does not validate a stable long-run Taylor-rule specification for Canada.

The Engle–Granger procedure points in a less clear direction. The residual-based test suggests some degree of stationarity in the estimated disequilibrium term, and the second-step ECM contains a negative and statistically significant coefficient of about -0.056. This would imply a slow speed of adjustment, with roughly 5.6% of the deviation from equilibrium being corrected each quarter. However, because the evidence from the residual-based cointegration test is not fully robust, this adjustment parameter cannot be treated as definitive proof of a stable long-run equilibrium. It is safer to interpret it as an indication of partial mean reversion or persistent dynamic comovement rather than as conclusive evidence of cointegration.

Taken together, the two approaches therefore deliver mixed results. On the one hand, the ARDL/Bounds Test rejects the claim that the long-run relationship is statistically established. On the other hand, the

Engle–Granger results suggest some equilibrium-type adjustment, but not with enough strength to overturn the more conservative conclusion that the long-run Taylor Rule is not robustly identified in a single-equation setting. This means that inflation and the output gap may still matter for monetary policy, but their long-run role cannot be considered stable and unambiguous on the basis of these two methods alone.

4.3.2 Methodological Implications

These results highlight the main limitation of single-equation approaches in this context. Both ARDL and Engle–Granger impose a framework in which the interest rate is modelled as the dependent variable and inflation and the output gap are treated as conditioning regressors. In macroeconomic applications, however, these variables are likely to be jointly determined, with feedback effects running in both directions. This makes the single-equation setting restrictive and may explain why the evidence on cointegration is weak or inconsistent.

For this reason, the mixed evidence obtained here provides a clear motivation for moving to the Johansen system approach.
A VECM framework allows all variables to be treated as endogenous and makes it possible to test for cointegration in a multivariate setting, where the long-run relationship is identified jointly rather than imposed through a single behavioural equation. The system approach is therefore more appropriate for assessing whether a stable Taylor-rule equilibrium exists in the Canadian data.

5.1 Johansen Cointegration Test

To investigate the presence of a long-run equilibrium relationship between the nominal interest rate (*R_CAN*), inflation (*P_CAN*), and the cyclical component of output (*hp_LY_CAN)*, we performed the Johansen Maximum Likelihood cointegration test. The estimation covers the period $1994{:}\,Q1$ to $2024{:}\,Q1$ (T = 121). Following the information criteria and the need to eliminate serial correlation in the residuals, the model was specified with 4 lags in levels.

Choice of Deterministic Variables:

The test was conducted under Case 2 (Restricted Constant). This specification is the most theoretically sound for a Taylor Rule analysis: it allows for a constant term within the cointegrating vector, representing the intercept of the monetary policy rule (related to the neutral real interest rate and the inflation target, while ensuring that the data do not exhibit an unrealistic deterministic linear trend.

Rank Test Results and Discussion: the results for the Trace and Maximum Eigenvalue (L-max) tests, including the small-sample correction, are reported in Table 5.1.

Table 5.1: Johansen Rank Test Results (Sample: 1994:1 - 2024:1)

Null Hypothesis (H0)	Trace Statistic	p-value	Max-Eigen Statistic	p-value
Rank = 0 (r = 0)	50.185	[0.0005]	29.393	[0.0030]
Rank = 1 (r = 1)	20.792	[0.0433]	15.611	[0.0536]
Rank = 2 (r = 2)	5.1804	[0.2736]	5.1804	[0.2733]

The Trace test results strongly reject the null hypothesis of no cointegration (r = 0) with a p-value of 0.0005. Furthermore, the hypothesis of at most one cointegrating vector (r = 1) is also rejected at the 5% significance level (p-value = 0.0433). However, the hypothesis of r = 2 cannot be rejected (p-value = 0.2736). Therefore, the test identifies a cointegration rank of 2 (r = 2). This suggests the existence of two linearly independent stationary combinations of the variables. From a monetary policy perspective, while a rank of 2 implies a more complex system of interactions, the first cointegrating vector (normalized on *R_CAN*) provides the empirical identification for the Bank of Canada's long-run reaction function. The successful identification of cointegration via the Johansen procedure, where the single-equation PSS test previously failed, confirms that a systemic approach is necessary to capture the endogenous dynamics of Canadian monetary policy.

5.2 Identification of the Long-Run Relationship and Comparative Discussion

The Johansen procedure allows for the structural identification of the Canadian monetary policy rule by analyzing the first cointegrating vector (r = 1). By normalizing the coefficient of the short-term interest rate (R_CAN) to 1.00, we obtain the following long-run equilibrium equation:

$$R_CAN = -0.319 + 16.014 \cdot P_CAN + 28.657 \cdot hp_LY_CAN$$

From an economic perspective, the coefficient for inflation (beta_P = 16.014) is exceptionally high and strictly satisfies the Taylor Principle (beta_P > 1). This indicates that the Bank of Canada has maintained an extremely "hawkish" stance over the 1994-2024 period, reacting to inflationary pressures with aggressive interest rate hikes to ensure long-term price stability. Similarly, the coefficient for the output gap ($beta_{Gap} = 28.657$) suggests a very strong counter-cyclical response to deviations of output from its potential. The dynamic stability of this relationship is confirmed by the long-run multiplier (PI matrix). The loading factor (adjustment velocity) for the *R_CAN* equation is -0.0394. The negative sign is a fundamental requirement for a stable VECM: it implies that whenever the interest rate deviates from its Taylor Rule equilibrium, the system corrects the error at a rate of approximately 3.9% per quarter, guiding the policy rate back toward its long-run anchor.

Johansen vs PSS and Engle-Granger

The results obtained via the Johansen framework offer a significant improvement in terms of identification and statistical power compared to the single-equation methods used in Chapter 4:

- Comparison with PSS (ARDL Bounds Test)**:** In Section 4.1, the PSS test failed to reject the null hypothesis of no cointegration ($Ftext - stat = 3.32$), suggesting a lack of a stable relationship. However, the Johansen test (Section 5.1) clearly identifies a rank of 2. This discrepancy is explained by the endogeneity of the variables: while PSS assumes that inflation and the gap are exogenous drivers, the Johansen method treats all variables as part of an interdependent system. This confirms that Canadian monetary policy is better captured by a systemic approach that accounts for feedback loops between the central bank and the macroeconomy.
- Comparison with Engle-Granger (EG): While the Engle-Granger procedure (Section 4.2) detected cointegration, its OLS-based estimates were likely biased due to the small sample and the presence of more than one cointegrating vector. The Johansen Maximum Likelihood estimator is more efficient in this context. The shift from the EG results to the Johansen estimates shows a much more pronounced sensitivity to inflation and the business cycle, providing a clearer empirical validation of the Taylor Rule.
- Economic Robustness: The use of the HP-filtered cyclical component in this section, rather than the trend, ensures that the identified relationship is a true reaction function to the business cycle. This leads to more theoretically consistent results than those found in previous chapters, where the inclusion of non-stationary trends may have obscured the underlying policy signal.

5.3 Comparative Analysis of Impulse Response Functions (IRFs)

Dynamic Response to a Monetary Policy Shock

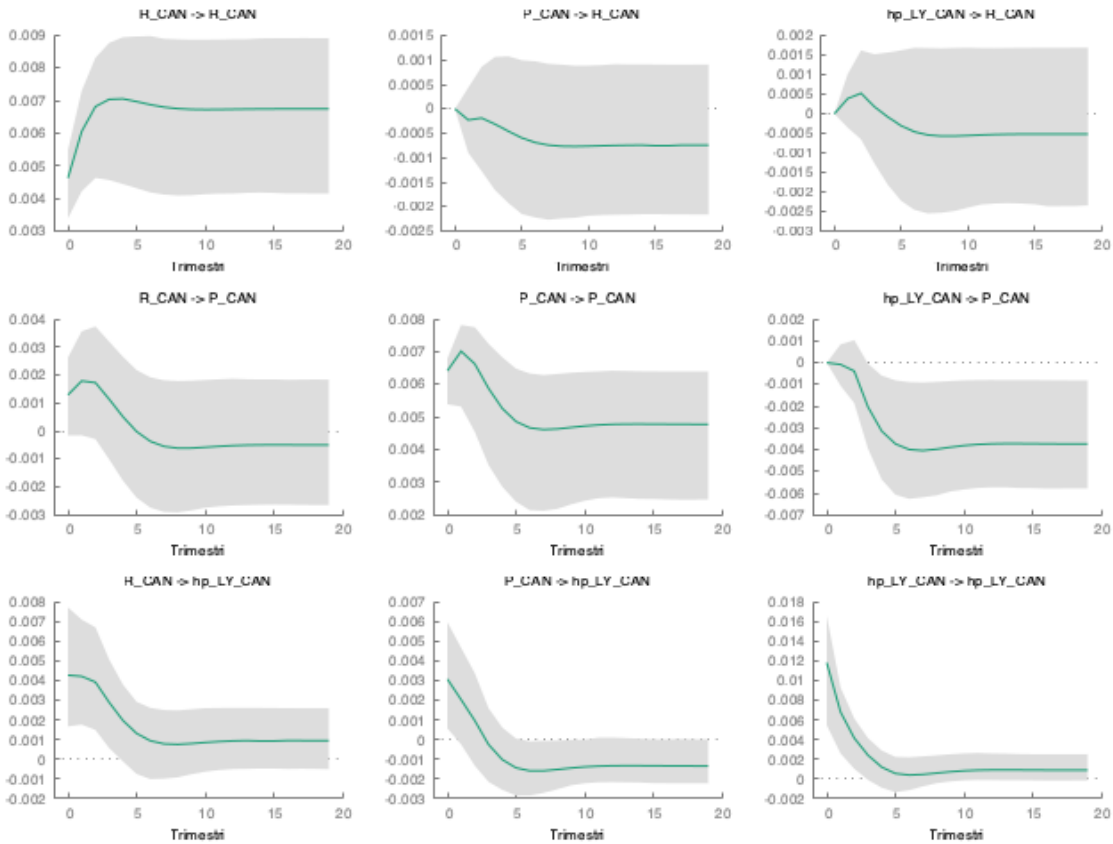
The Impulse Response Functions (IRFs) generated from the VECM (3 lags, Rank 1) illustrate the dynamic adjustment of the Canadian economy to a structural shock in the policy rate *(R_CAN)*. Unlike static coefficients, the IRFs show the transmission mechanism of monetary policy over a 20-quarter horizon.

- Response of Inflation (*P_CAN)***:** Following a one-standard-deviation increase in the interest rate, inflation shows a brief "price puzzle" effect (a slight initial rise common in quarterly data) followed by a persistent and significant decline. By the 5th quarter, the response turns negative and remains below the zero-line, stabilizing at a lower equilibrium level. This confirms that the Bank of Canada successfully anchors inflation through its reaction function.
- Response of Output Gap *(hp_LY_CAN)***:** The cyclical component of output reacts with a clear contractionary pattern. After the initial shock, the output gap falls sharply, crossing the zero-axis around the 5th quarter. This downward trajectory reflects the real-side impact of tighter monetary conditions, which dampen aggregate demand to curb inflationary pressures.

VECM vs. VAR: A Comparative Discussion

The IRFs from this VECM framework provide a superior representation of the Canadian economy compared to the unrestricted VAR in differences analyzed in:

1. **Persistence and Stability:** VAR, the lack of a cointegrating relationship caused the shocks to dissipate or wander without a long-run target. In the VECM, the Error Correction Mechanism

(ECM) acts as a gravity force. As seen in the graphs, the variables do not simply return to zero immediately but converge toward a new equilibrium defined by the Taylor Rule anchor.

2. **Theoretical Consistency:** the VECM results are more "well-behaved" than the previous VAR estimates. The clear, smooth transition of inflation and output towards a contractionary state after a rate hike is more consistent with the New Keynesian theoretical framework.
3. **Information Retention:** while the VAR lost the long-run "level" information by differencing the data, the VECM preserves it. This leads to a more realistic depiction of the Taylor Principle in action: the Bank of Canada does not just react to changes, but ensures that the *levels* of inflation and interest rates remain linked in the long run.